C1 Four Topology MOSbius

Schematic Review Slides 07/03

LINK to Video

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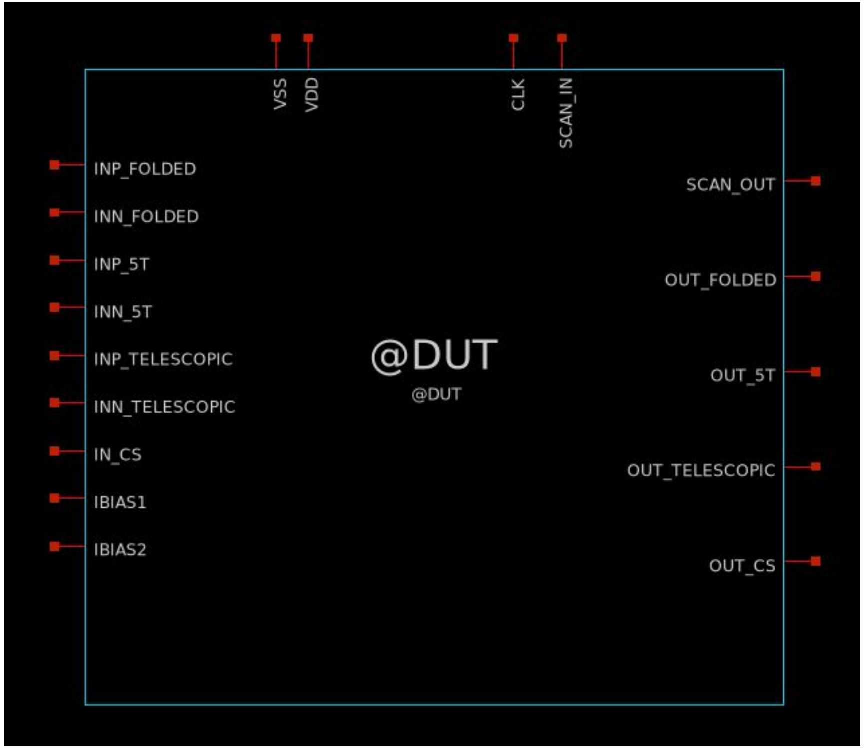
Project Information

Goal: To design and fabricate a chip containing several of the most common amplifier topologies, all with accessible pins, so that a student can wire up, test, characterize, and compare each amplifier. Write a short lab manual to accompany the chip (consider during design, but can follow design/layout submission). Students measure DC gain, GBW, slew rate, stability, PSRR, etc. for each amplifier, learning the trade-offs through hands-on experimentation

System Overview



Total Pins: 17

Global Input (2): VDD, VSS

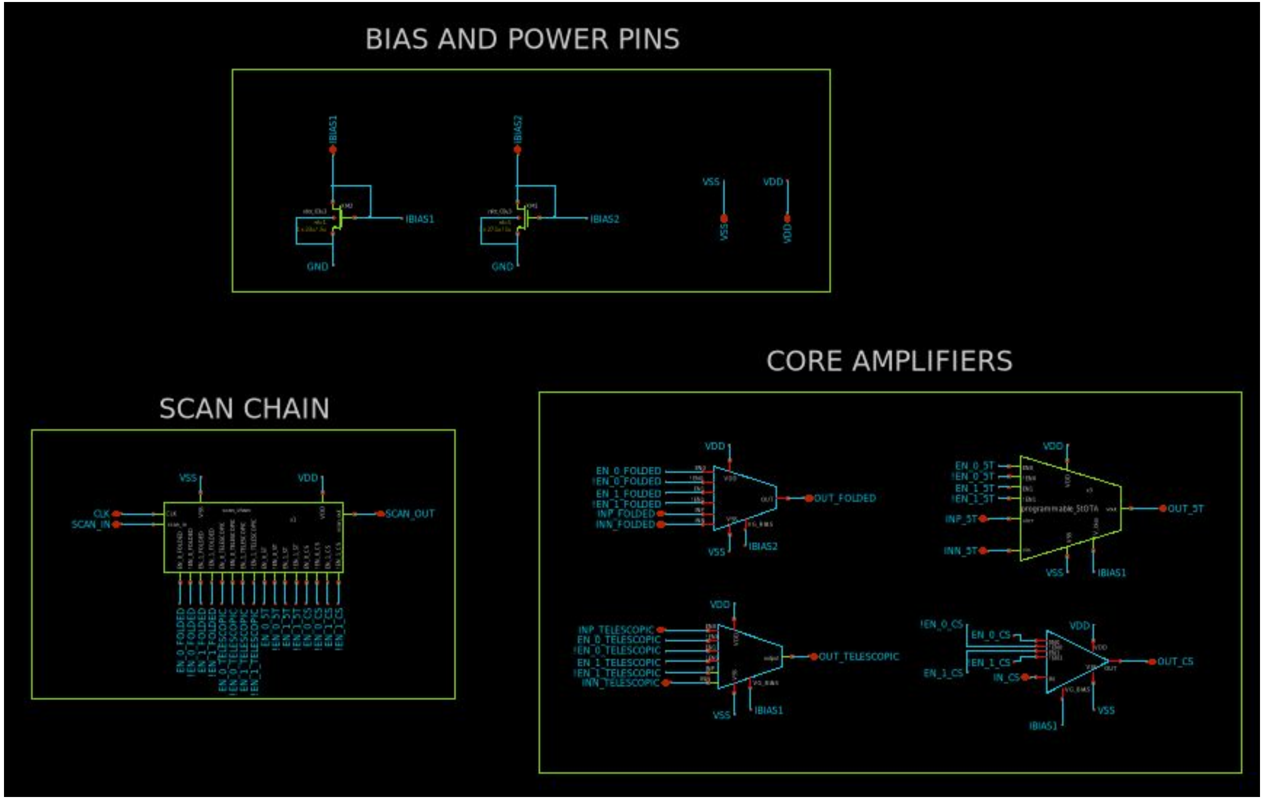
Digital Input (2): Scan In, Clock

Digital Output (1): Scan Out

Analog Input (8): IREF, IN(CS), INP/INN (5T), INP/INN (Telescopic), INP/INN (Folded)

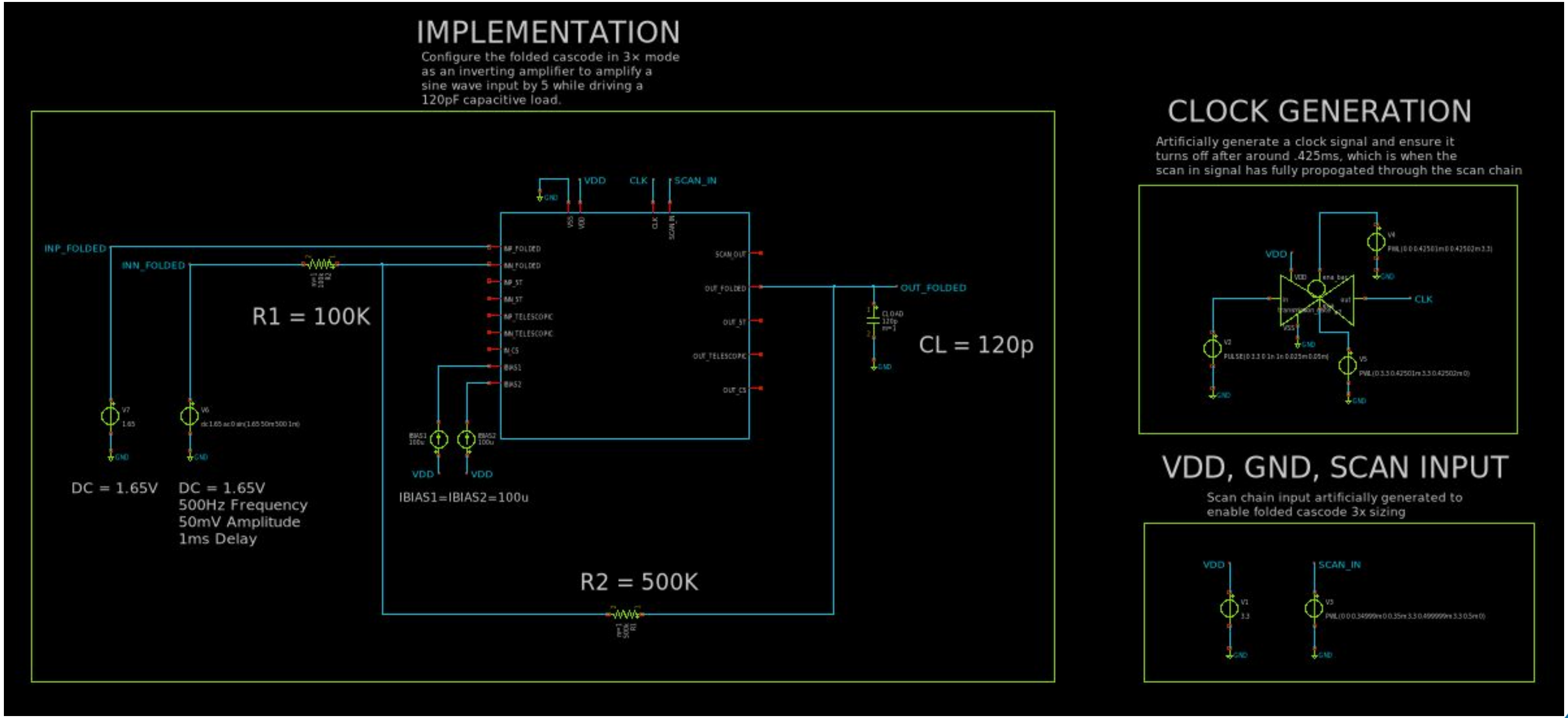
Analog Output (4): OUT (CS), OUT (5T), OUT (Telescopic), OUT (Folded)

System Block Diagram



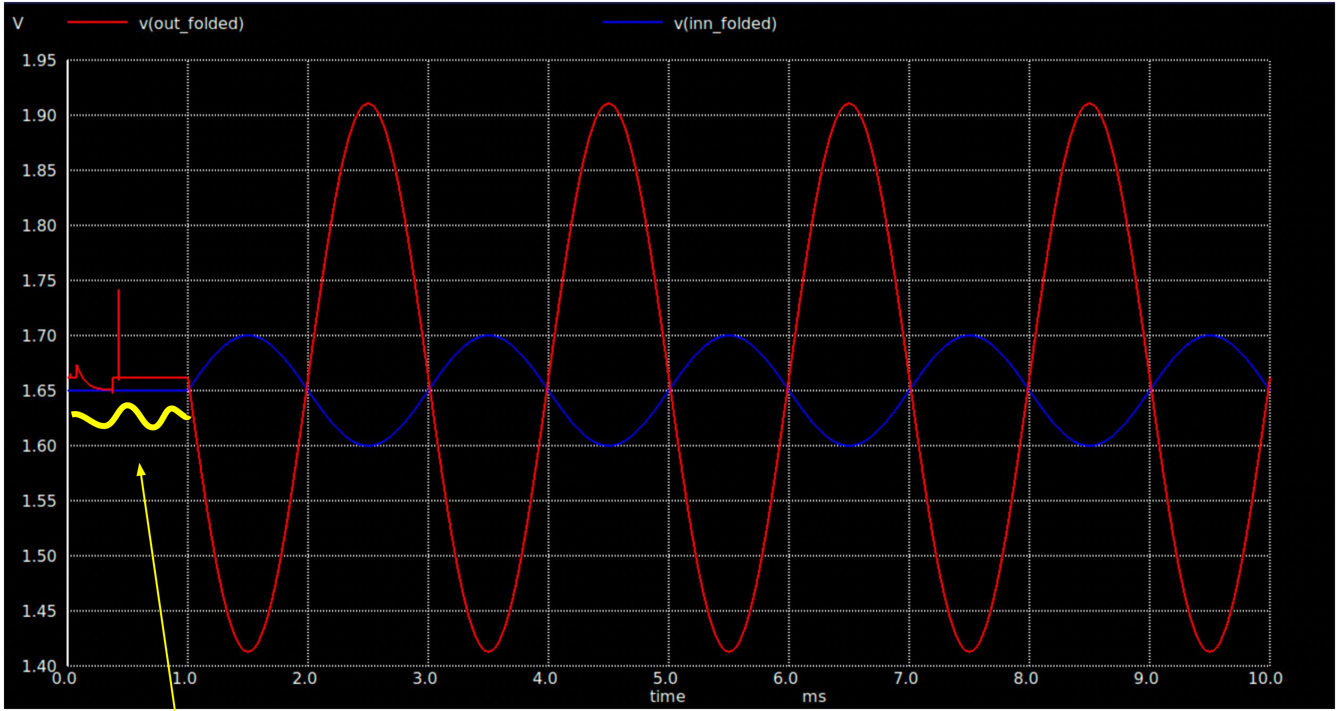
In practice, we will not have two separate bias currents. At the moment, there is some mismatch between the initial current mirrors in the amplifiers. This will be fixed so that we use only one externally generated bias current for the entire chip.

System Testbench



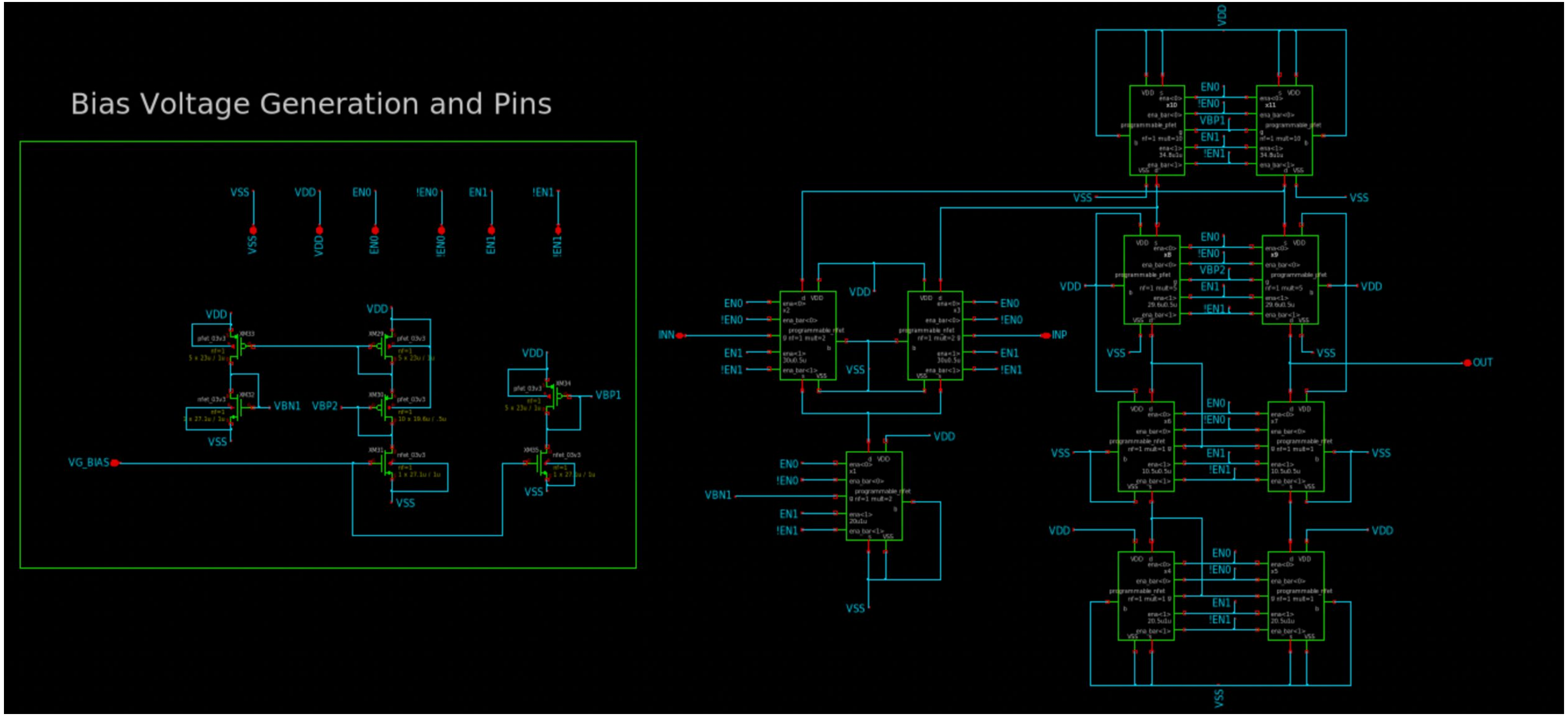
System Simulation Results

Measured Gain of 4.975 (Targeting 5)



Start Up (Scan In Propagating Through Scan Chain)

Programmable Folded Cascode Schematic

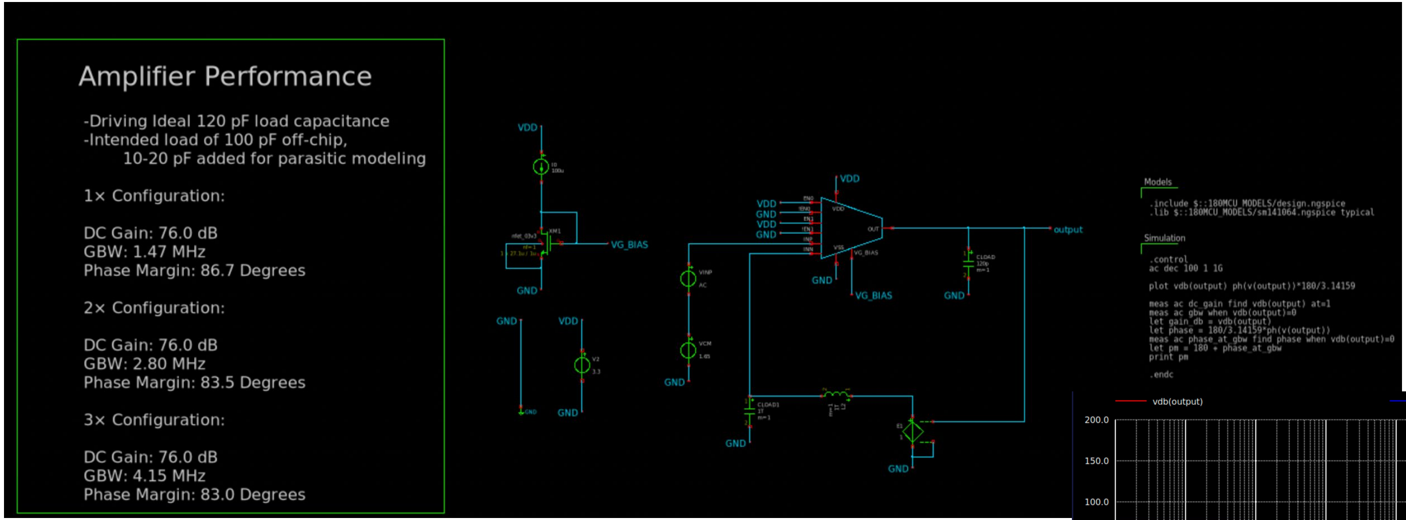


Sizing and Device Information

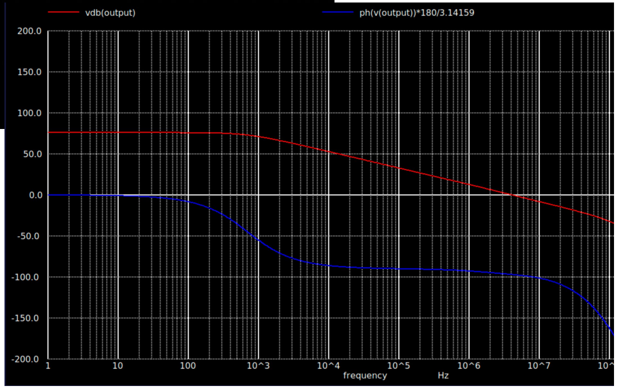
Device	Role	W (µm)	L (µm)	ID (µA)
XM10/XM11	NMOS input pair	60	0.5	75
XM12	NMOS tail	40	1	150
XM2/XM5	PMOS current sources	174	1	150
XM3/XM4	PMOS cascodes	148	0.5	75
XM6/XM7	NMOS cascodes	10.5	0.5	75
XM8/XM9	NMOS mirrors	20.5	1	75

- VDD = 3.3 V
- IBIAS = 100 uA
- 3.3 V Devices
- Sizing for the original amplifier and base sizing for programmable amplifier is listed in the table
- Total current consumption is approximately 300uA, giving roughly 1mW at 3.3V

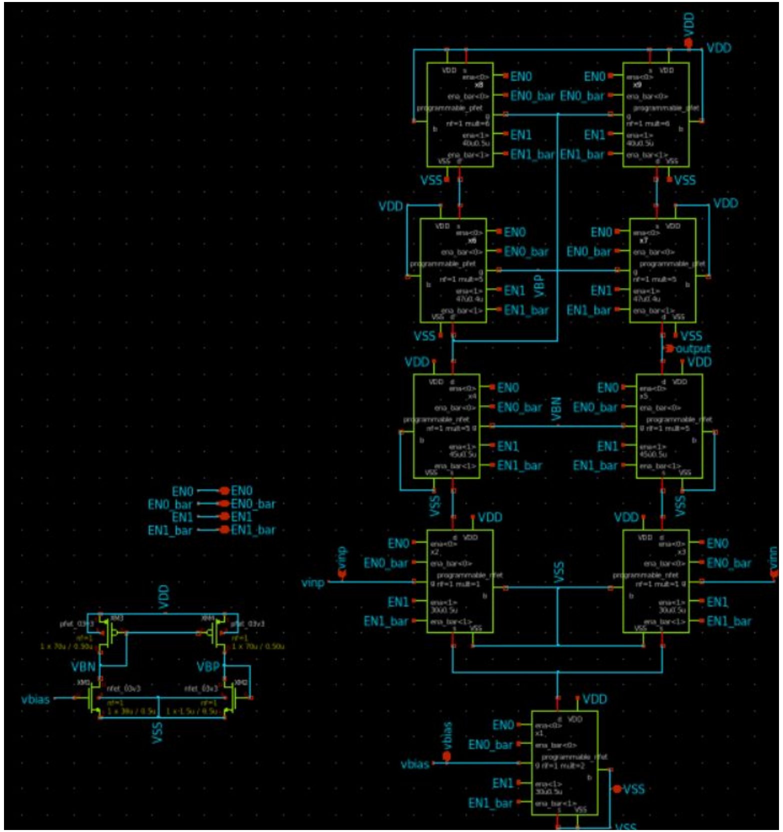
AC Open-Loop



Simulation
Result from 3x

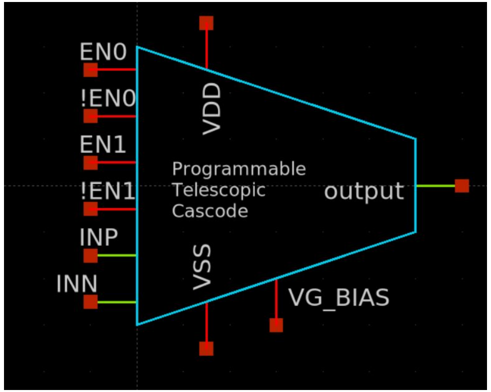


Programmable Telescopic Cascode

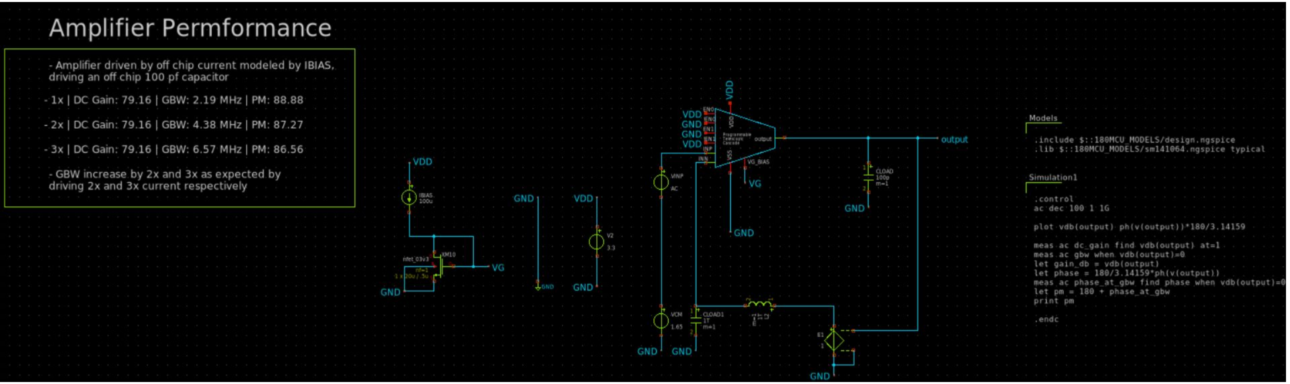


ID	Programmable | MOSFET	W (μm) (1x)	L (μm) (1x)	Multiplier
X1	Programmable	30	.5	2
X2, X3	Programmable	30	.5	1
X4, X5	Programmable	30	.5	6
X6, X7	Programmable	47	.4	5
X8, X9	Programmable	30	.5	8
XM3, XM4	PMOS	70	.5	1
XM2	NMOS	1.5	.5	1
XM1	NMOS	30	.5	1

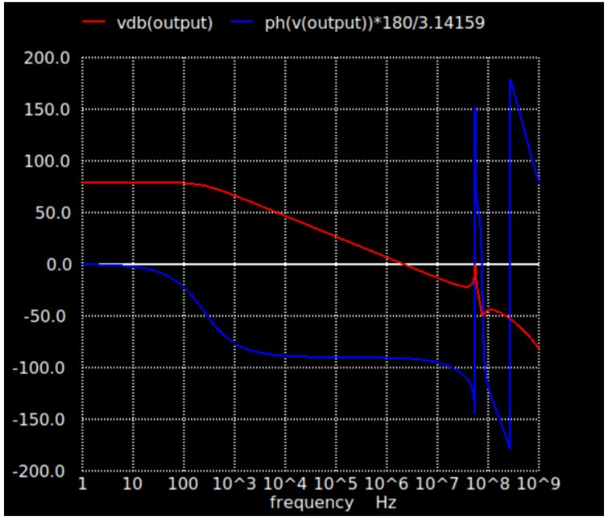
Symbol View



AC Open Loop

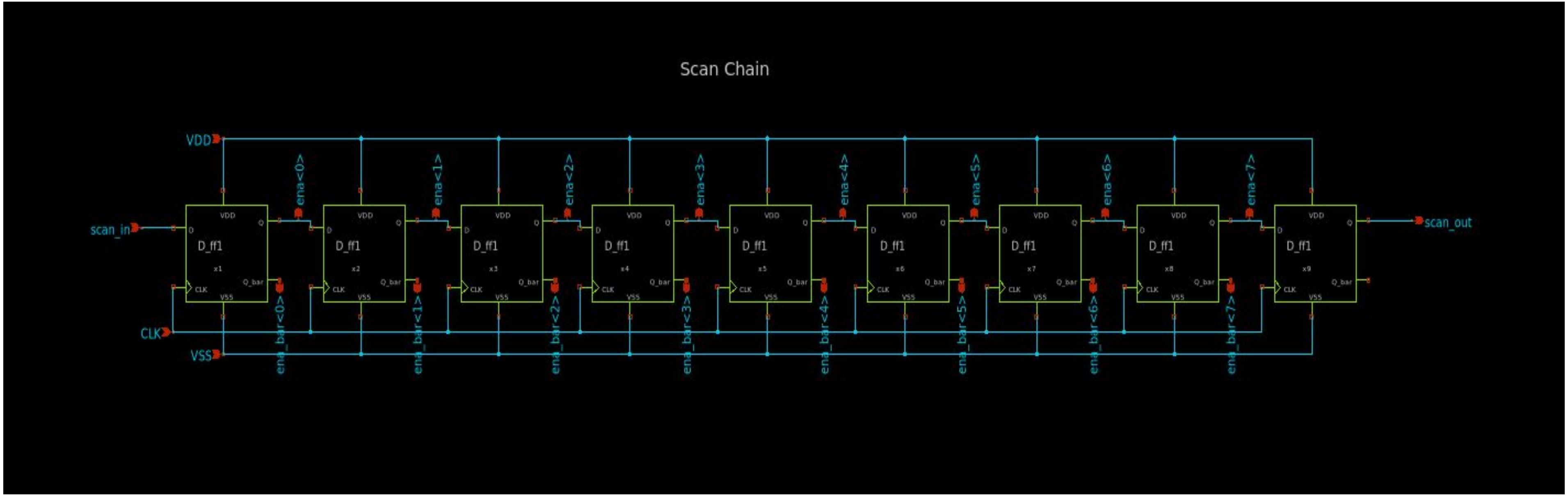


1X Gain and GBW match that of non-programmable telescopic cascode found in extended slide deck



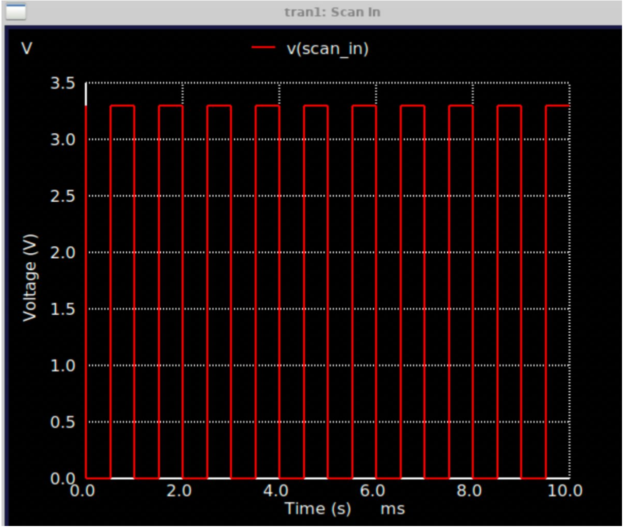
1X Bode Plot

Scan Chain

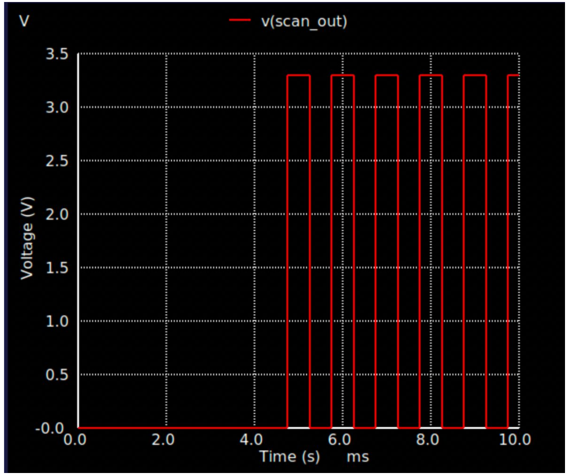


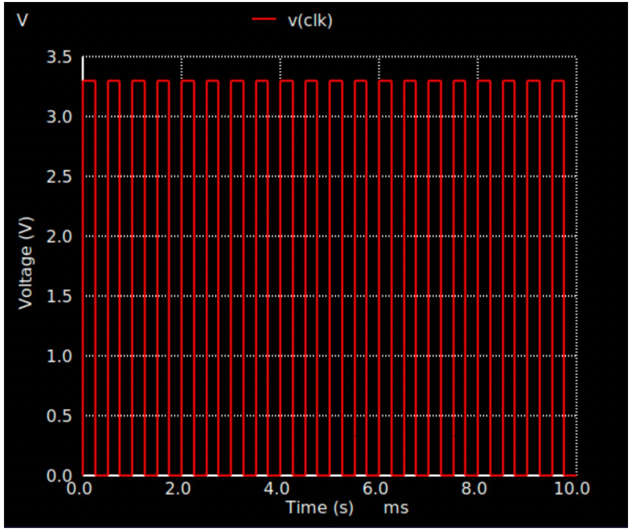
Scan Chain
VDD
scan_in
CLK
VSS
scan_out
D_ff1
ena<0>
ena<1>
ena<2>
ena<3>
ena<4>
ena<5>
ena<6>
ena<7>
ena_bar<0>
ena_bar<1>
ena_bar<2>
ena_bar<3>
ena_bar<4>
ena_bar<5>
ena_bar<6>
ena_bar<7>

Scan Chain Simulation



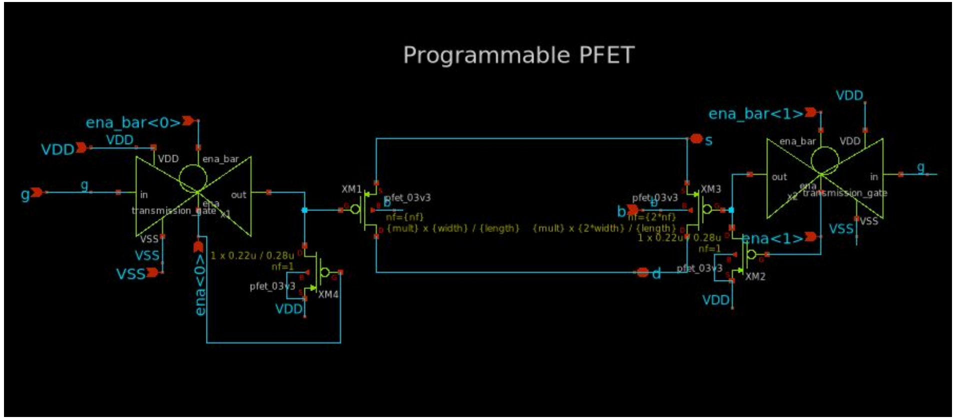
- Scan out does not match scan in until 9 clock cycles due to there being 9 flip flops





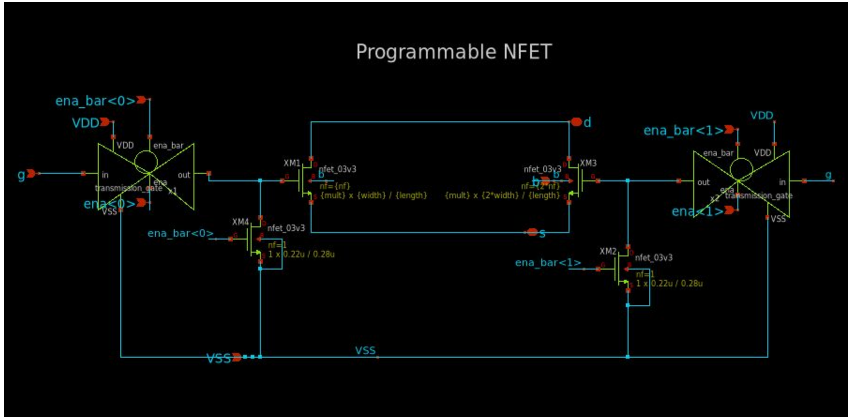
- Freq = 2kHz
- Has been tested and working at 20 kHz

Programmable Transistors

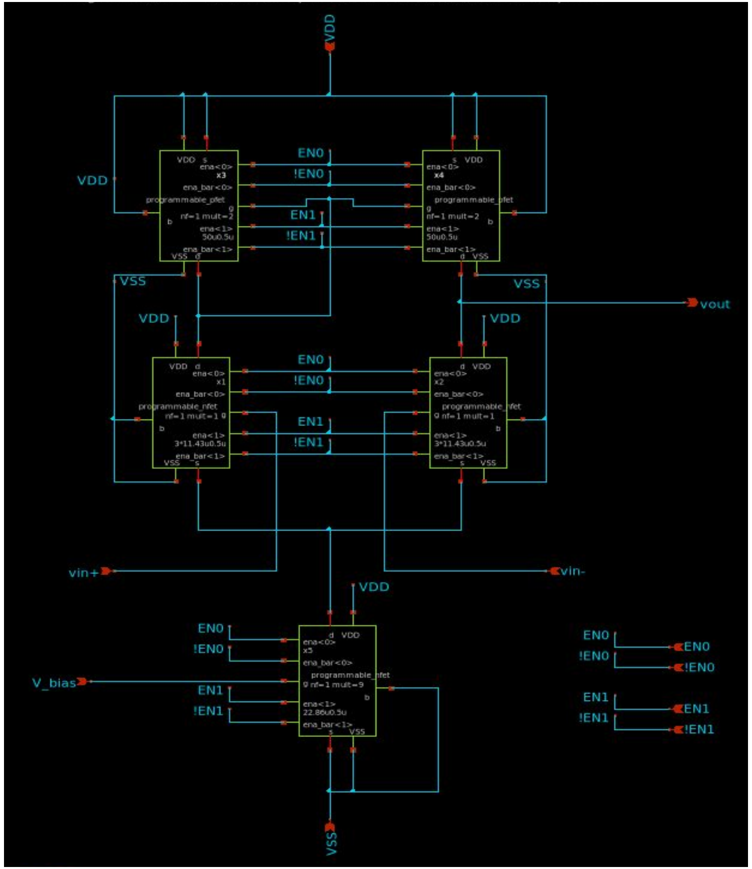


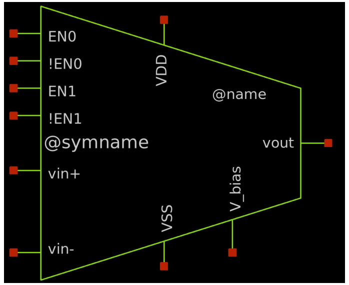
Pull Up transistors to avoid floating gates

Pull Down transistors to avoid floating gates



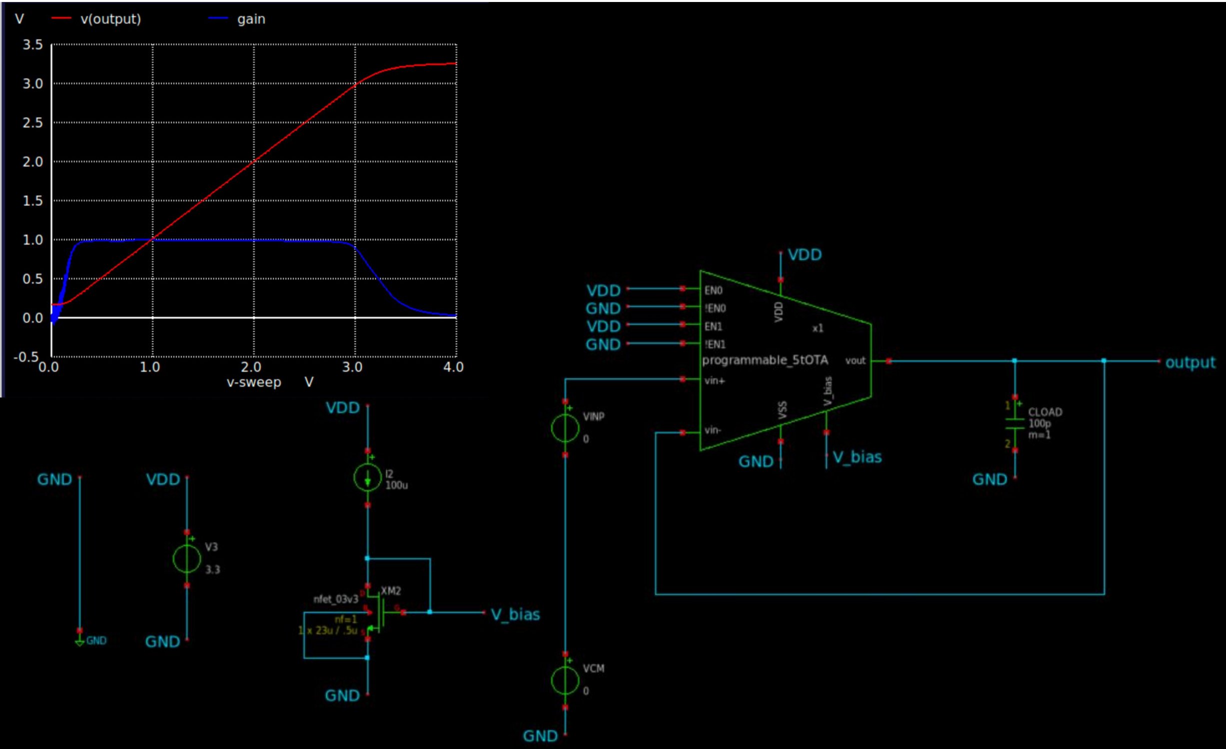
Programmable 5-Transistor OTA





ID	Programmable | MOSFET	W (µm) (1x)	L (µm) (1x)	Multiplier
X3, X4	Programmable (PFET)	50	.5	2
X1, X2	Programmable (NFET)	11.43	.5	1
X5	Programmable (NFET)	22.86	.5	9

Programmable 5-Transistor OTA - DC Unity



Amplifier Performance(DC)

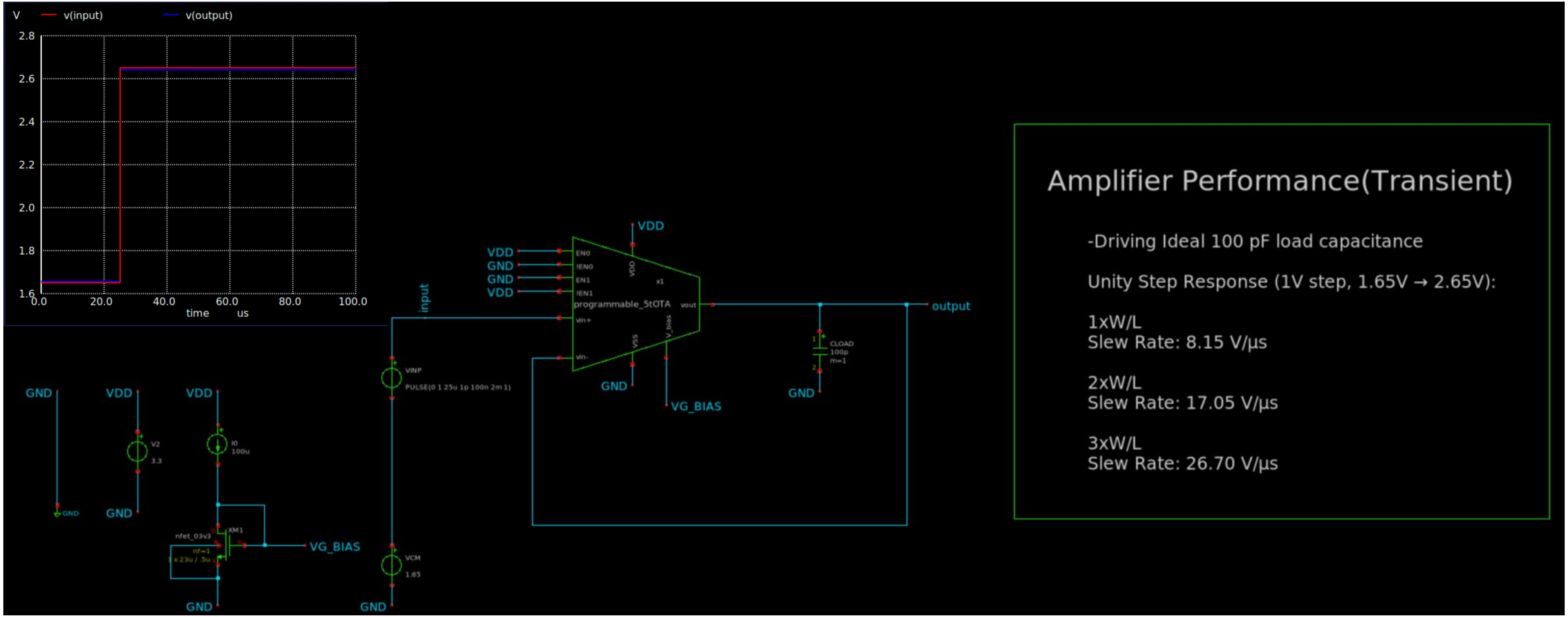
-Driving Ideal 100 pF load capacitance

1xW/L
ICMR: 0.298V – 2.924V (2.626V range)
Output Swing: 0.298V – 2.924V (2.626V range)

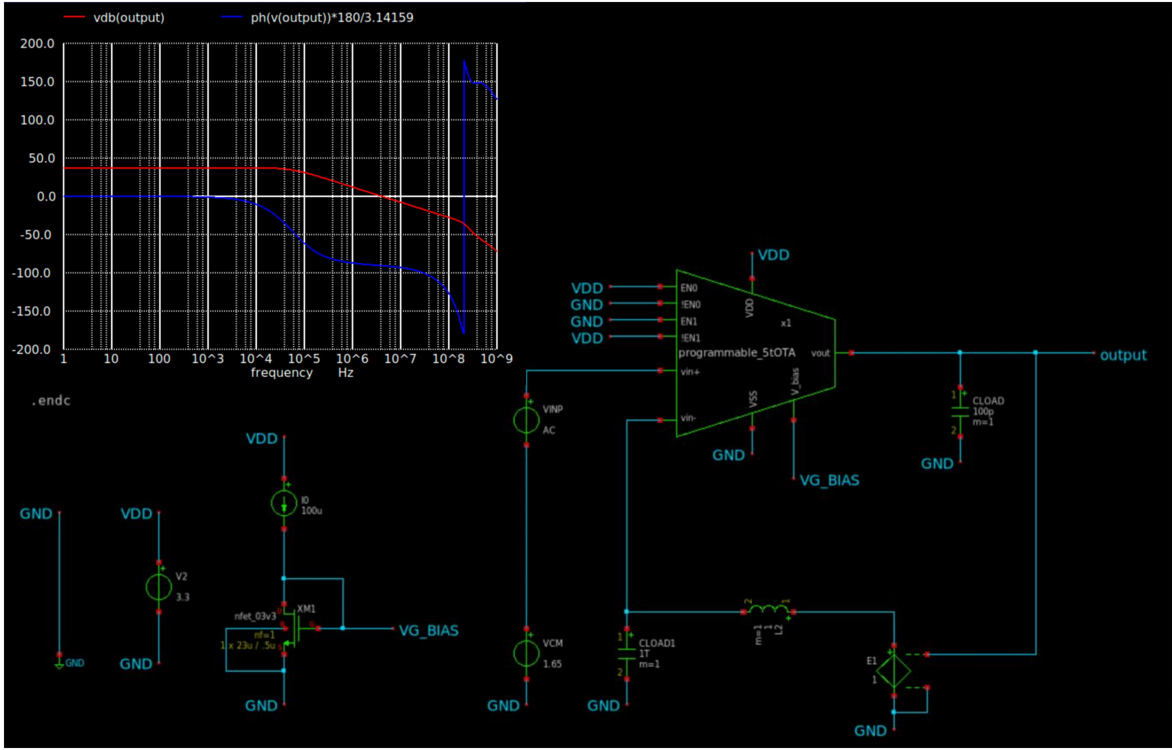
2xW/L
ICMR: 0.271V – 2.924V (2.653V range)
Output Swing: 0.271V – 2.924V (2.653V range)

3xW/L
ICMR: 0.261V – 2.924V (2.663V range)
Output Swing: 0.261V – 2.924V (2.663V range)

Programmable 5-Transistor OTA - Slew Rate



Programmable 5-Transistor OTA AC



Amplifier Performance(AC)

-Driving Ideal 100 pF load capacitance

1xW/L
DC Gain: 37.5 dB
3-dB BW: 54.8 kHz
GBW: 4.12 MHz
PM: ~90.0 Degrees

2xW/L
DC Gain: 37.5 dB
3-dB BW: 107.3 kHz
GBW: 8.08 MHz
PM: ~90.0 Degrees

3xW/L
DC Gain: 37.5 dB
3-dB BW: 159.8 kHz
GBW: 12.0 MHz
PM: ~90.0 Degrees

Next Steps

- Continue to Test System-Level Configurations and Specifications
- Experiment and Document Two-Stage Configurations, Compensation
- More Characterization Simulations (Noise and PSRR)
- Begin Layout

Thank You!

Any feedback is greatly appreciated. Please view our full slide deck linked on our repository and issue for more simulations and calculations.